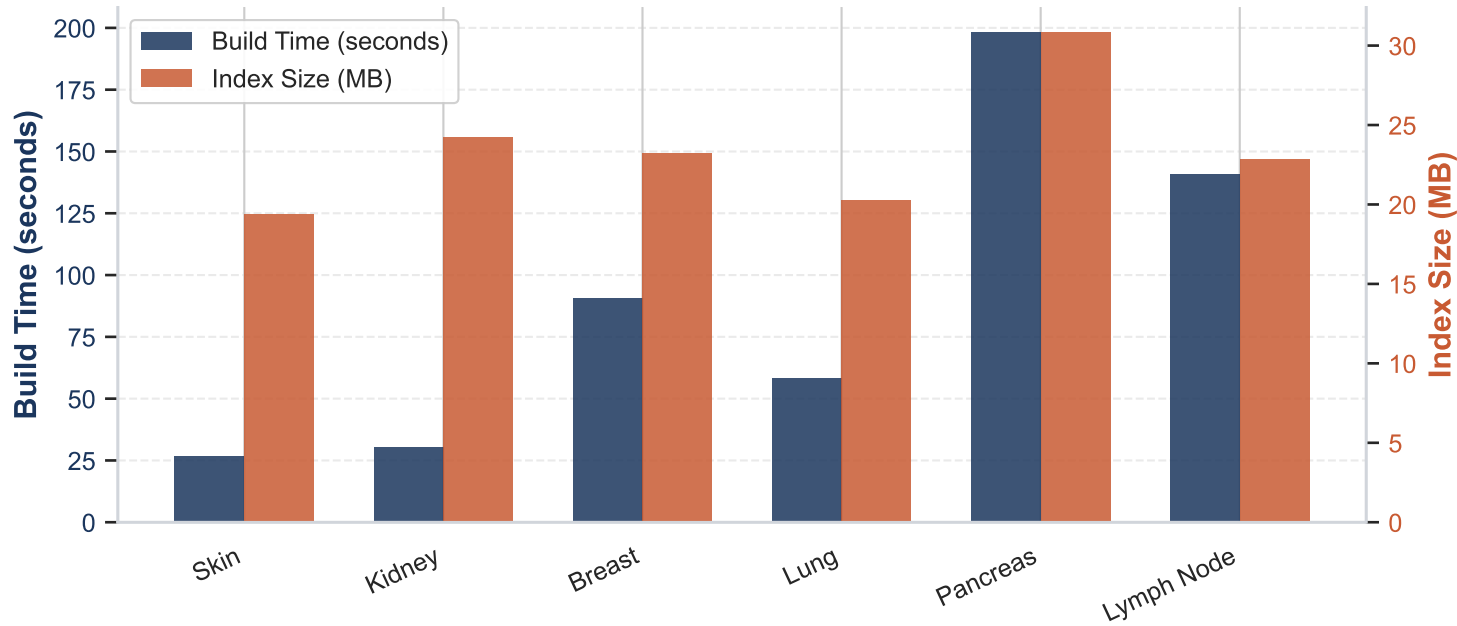


Figure 1: SPINDLE Core Architecture, Ultra-Fast Indexing, and Spatial Search Performance

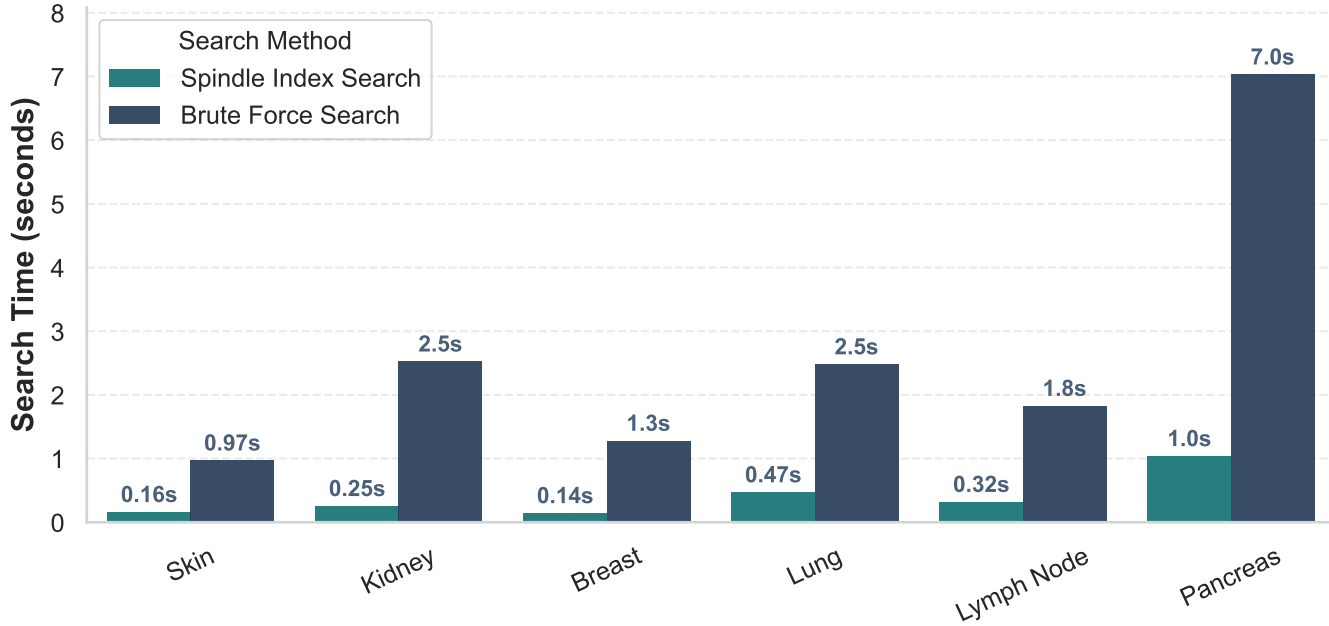
A

Index Scalability and Memory Footprint of Spindle
(Sub-linear Build Time & Compact Storage across Datasets)

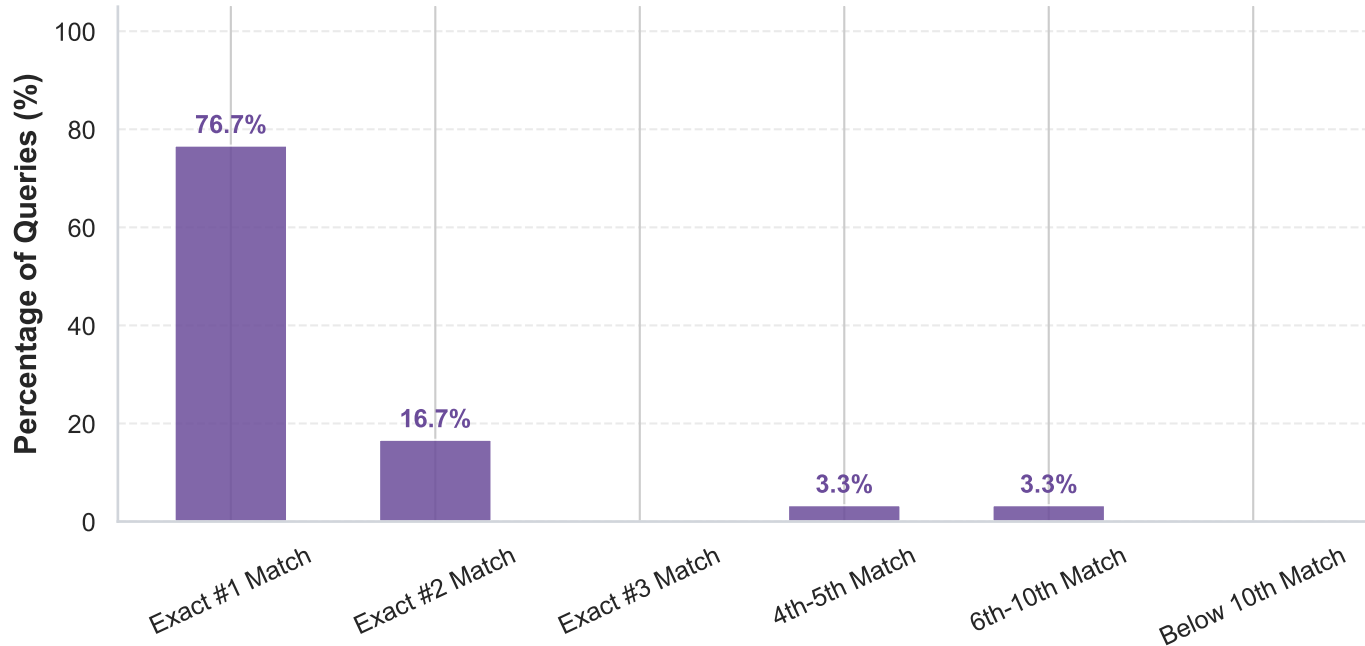


B

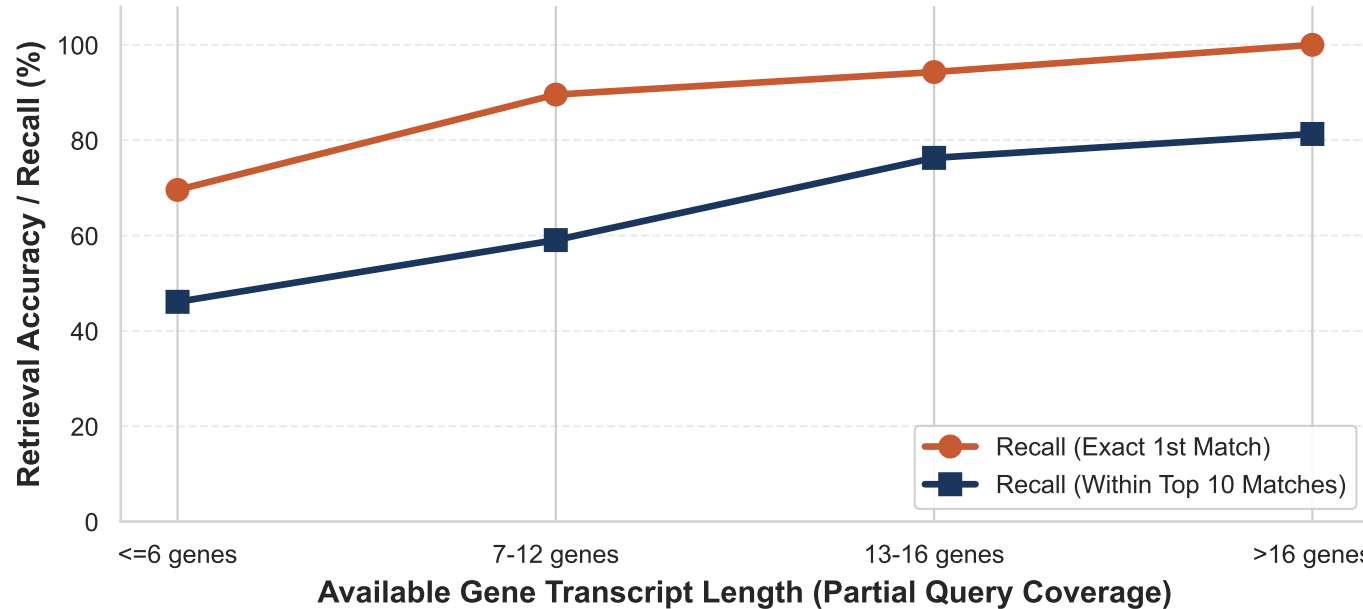
Spindle Index Search Time compared to Brute Force Search
(Exact Query Execution Time in Seconds Across Datasets)



Accuracy of Spindle First Match
(Percentage of queries where Spindle found the exact #1 best match, #2 best match, and so on)

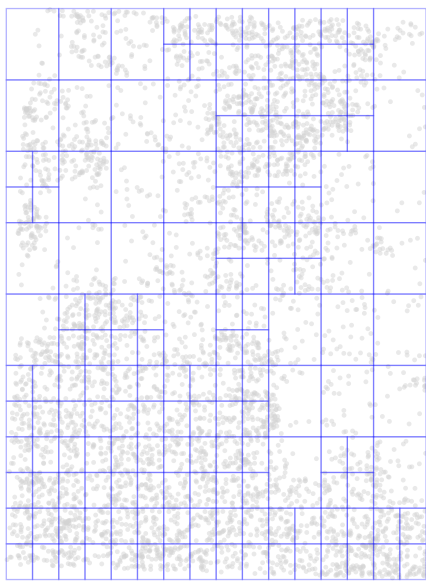


Partial Query Search Accuracy
(High recall despite searching with incomplete or truncated gene transcripts)

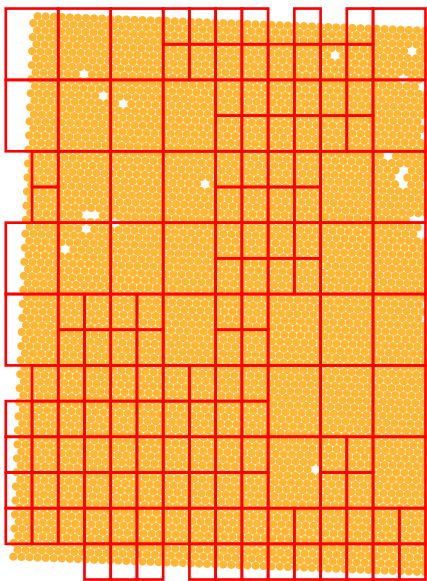


E

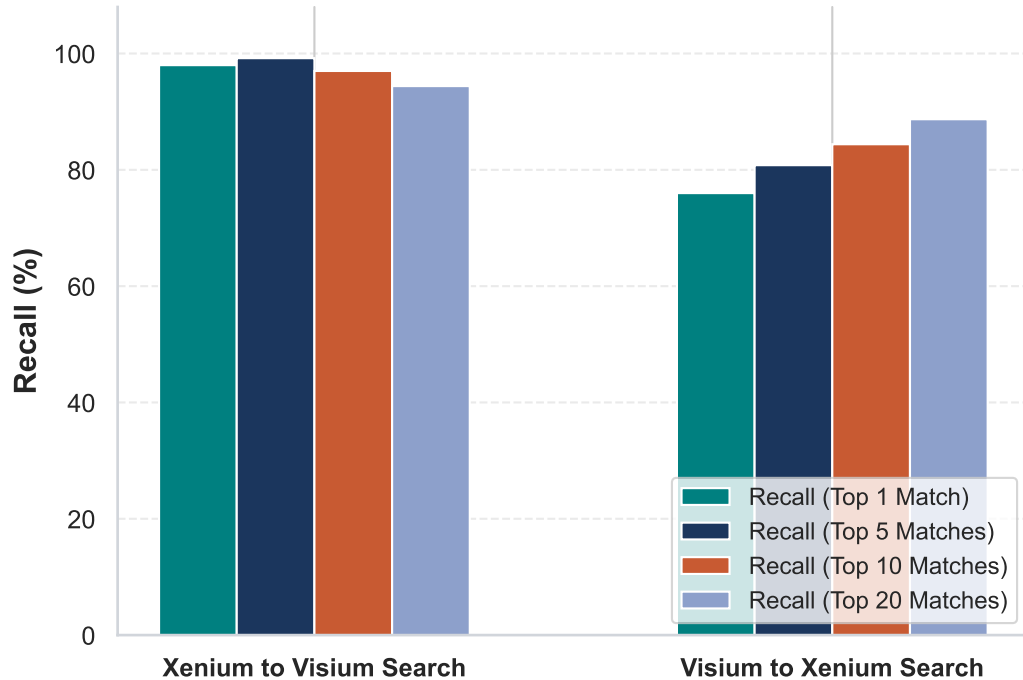
Xenium (166 tiles)



Visium (157 tiles)

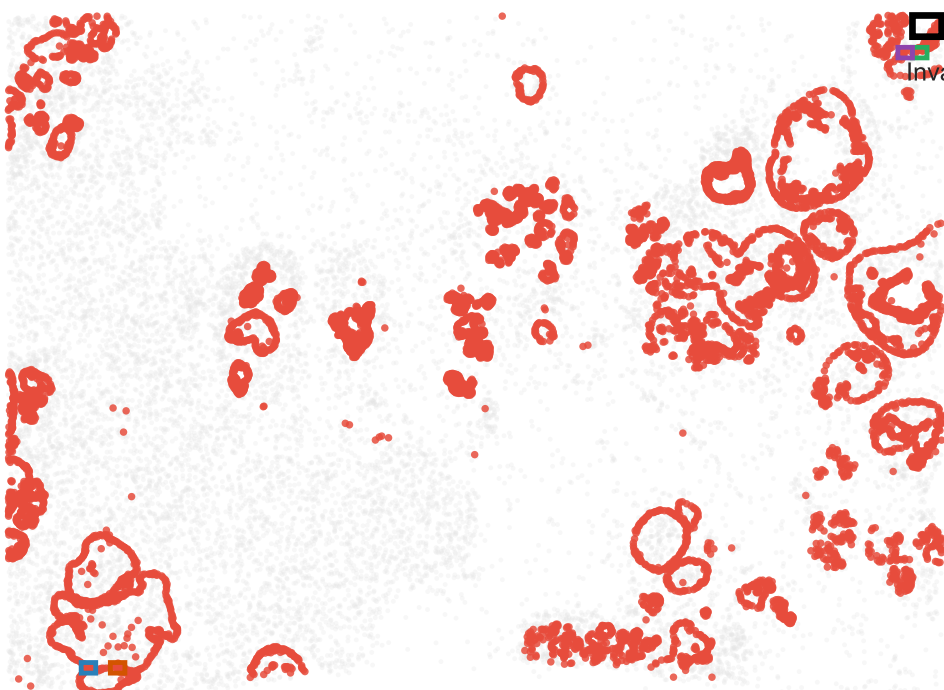


Cross-Modality Retrieval Accuracy
(Percentage of true matches found within top candidates)



F

Target Niche Spatial Discovery Map
(Retrieved Basal Layer Spatial Niche)



Target Cell Abundance: Background vs. Top 50 vs. Top 10
(Spatial Niche Retrieval Enrichment)

